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Super Composition Algebras and the Standard Model

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Abstract

We introduce and describe super-composition algebras and devise a way to get the Standard Model bosons and fermions unified into a single super-Lie algebra that acts as a symmetry for the super-octonions. We discuss the dynamics of the obtained action and how it deviates from the Standard Model in the gravity sector. We further exploit the algebraic structures to speculate the existence of three generations of fermions from the super-triality symmetry of the super-octonions. Finally, we discuss some interesting mathematical observations relating twistors to split-octonions.

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1 Background

Recent advances in experimental particle physics seem to suggest that there are no new fundamental particles in the Standard Model upto the electroweak scale. Other than the neutrino sector it is very improbable to find new fundamental particles at our energy scales. This makes it the perfect time for us to understand the Standard Model as it is from a more fundamental mathematical background. The purpose of this project is to understand the Standard Model content from an underlying algebra and to be able to answer some fundamental questions like why there should be just three generations of fermions.

A famous theorem says that there are only four division algebras, the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$, the quaternions $\mathbb{H}$, and the octonions $\mathbb{O}$ [1, 2]. In mathematics, these algebras are closely related to the classification of compact, simple Lie algebras and Euclidean Jordan algebras. They are also related to Clifford algebras and spin structures in different spacetime dimensions. Spacetimes in different superstring theories can be understood as the vector bundle of these division algebras over the string worldsheet. Many people have tried to construct the Standard Model fermion content using these

division algebras with the gauge bosons as Lie algebra-valued fields acting on them.

In this project, we try to unify both the bosonic gauge sector and the fermionic spinor sector of the Standard Model into a single superalgebra. The $\mathbb{Z}_2$ grading in the superalgebra is over the space of fermions and bosons. To do so, we make use of the super-composition algebras which are defined below.

2 Super-Composition Algebras

It is through logic that we prove, but through intuition that we discover.

– Henri Poincaré

A composition algebra $\mathbb{A}$ over a field $\mathbb{F}$ is equipped with a symmetric bilinear inner product $\langle _ | _ \rangle : \mathbb{A} \times \mathbb{A} \rightarrow \mathbb{F}$ such that:

$$\langle xy | xy \rangle = \langle x | x \rangle \langle y | y \rangle \quad \forall x, y \in \mathbb{A}. \quad (1)$$

There are only 4 composition algebras over the field of complex numbers namely the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$, the quaternions $\mathbb{H}$, and the octonions $\mathbb{O}$. Over the field of real numbers, there are only 7 composition algebras, namely the 4 division algebras that have an anisotropic inner-product and their split-versions with an isotropic split inner-product, the split-complex numbers $\mathbb{C}'$, the split-quaternions $\mathbb{H}'$, and the split-octonions $\mathbb{O}'$. In total, these 11 algebras are the only composition algebras.

One can define the norm of the composition algebras as $N(x) = \langle x | x \rangle$. It then immediately follows from the composition law that:

$$\langle x | y \rangle = \frac{1}{2}(N(x + y) - N(x) - N(y)). \quad (2)$$

One can polarize either $x \rightarrow x + u$ or $y \rightarrow y + v$ in the composition law (1) to obtain what are called the scaling laws:

$$\langle xy | uy \rangle = \langle x | u \rangle \langle y | y \rangle, \quad (3)$$

$$\langle xy | xv \rangle = \langle x | x \rangle \langle y | v \rangle. \quad (4)$$

Further, one can polarize both arguments of the inner-product in the composition law to obtain what is called the exchange law:

$$\langle uv | wx \rangle = 2\langle u | w \rangle \langle v | x \rangle - \langle ux | wv \rangle. \quad (5)$$

A superalgebra is an algebra equipped with a $\mathbb{Z}_2$ grading such that we can divide our algebra into two parts, namely a bosonic and fermionic subalgebra. The rule of the game is that the multiplication of any element in the algebra with a boson does not change its grading whereas multiplying it by a fermion will change it from a boson to a fermion or the other way round. This can be expressed mathematically as:

$$A = A_0 \oplus A_1, \quad A_i A_j \in A_{i+j \pmod{2}}. \quad (6)$$

A_0 and A_1 are the bosonic and fermionic subspaces, respectively. $|A|$ is either 0 (for bosons) or 1 (for fermions). Such an algebra is called *super-commutative* (like a Jordan superalgebra) if the super-commutator vanishes

$$[a, b] \equiv ab - (-1)^{|a||b|}ba = 0 \quad \forall a, b \in A \quad (7)$$

and *super-anticommutative* (like a Lie superalgebra) if the super-anticommutator vanishes

$$\{a, b\} \equiv ab + (-1)^{|a||b|}ba = 0 \quad \forall a, b \in A. \quad (8)$$

Relaxing our notation henceforth, minus one raised to the power something in the algebra $(-1)^x$ is actually minus one raised to the power its mod $(-1)^{|x|}$.

It is easy to see that the endomorphisms (linear transformations) of a super-vector space form a superalgebra, moreover, they also create a supergroup. A supergroup is a $\mathbb{Z}_2$ graded group that needs to have an identity and inverse.

A superalgebra can be defined over the conventional field of real numbers or complex numbers as is usually done in linear algebra. Kac and Kantor worked with these fields to classify super-Lie and super-Jordan algebras [3, 4]. Another possibility is to work with supernumbers, i.e., Grassmann-valued numbers used by physicists to describe super-Poincare algebra and to quantize fermions that have anti-commuting statistics. A nice mathematical treatment of working with supernumbers can be found here [5, 6].

The complex supernumbers $\tilde{\mathbb{C}}$ (denoted Λ_∞ by DeWitt in [5]) are numbers of the form:

$$z = \sum_{n=0}^{\infty} \frac{1}{n!} c_{a_1 \dots a_n} \zeta^{a_1} \dots \zeta^{a_n} \quad (9)$$

where ζ^a ($a = 1, 2, \dots$) are an infinite set of independent anti-commuting Grassmann generators satisfying $\{\zeta^a, \zeta^b\} = 0$, and the coefficients $c_{a_1 \dots a_n}$ are $\mathbb{C}$ -valued and completely anti-symmetric under index permutation. The leading ($n = 0$) term in the expansion (9) is called the “body” $z_B \in \mathbb{C}$, while the remaining ($n \geq 1$) terms are called the “soul” z_S . The supernumber z has an inverse if and only if its body $z_B \in \mathbb{C}$ is non-zero, in which case the inverse is unique, and given by

$$z^{-1} = z_B^{-1} \sum_{n=0}^{\infty} (-z_B^{-1} z_S)^n. \quad (10)$$

We can also uniquely decompose z into its c-number (commuting) and a-number (anti-commuting) parts, which include an even or odd number of Grassmann generators, respectively:

$$z = z_c + z_a. \quad (11)$$

Now, given z , we can consider either its complex conjugate

$$\bar{z} = \sum_{n=0}^{\infty} \frac{1}{n!} \bar{c}_{a_1 \dots a_n} \zeta^{a_1} \dots \zeta^{a_n}. \quad (12a)$$

or its fermion conjugate

$$z' = z_c - z_a. \quad (12b)$$

Note that our convention for complex conjugation is different than DeWitt's: In our convention, the Grassmann generators are unaffected by complex conjugation, $\overline{\zeta^{a_1} \dots \zeta^{a_n}} = \zeta^{a_1} \dots \zeta^{a_n}$, so a supernumber z is real ($z = \bar{z}$) or imaginary ($z = -\bar{z}$) iff its coefficients $c_{a_1 \dots a_n}$ are all real or all imaginary, respectively.

The two conjugations (12a) and (12b) both define automorphisms (or super-antiautomorphisms) $\tilde{\mathbb{C}} \rightarrow \tilde{\mathbb{C}}$ which are involutions (*i.e.* which square to the identity):

$$\overline{(zw)} = \bar{z}\bar{w} = (-1)^{zw}\bar{w}\bar{z} \quad (13a)$$

$$(zw)' = z'w' = (-1)^{zw}w'z'. \quad (13b)$$

Complex conjugation splits $\tilde{\mathbb{C}}$, into real elements (with $\bar{z} = +z$) and imaginary elements (with $\bar{z} = -z$), while fermionic conjugation splits $\tilde{\mathbb{C}}$ into c -numbers (with $z' = +z$) and a -numbers (with $z' = -z$).

Within $\tilde{\mathbb{C}}$, the subalgebras fixed by complex conjugation ($z = \bar{z}$), by fermionic conjugation ($z = z'$), or both, are the real supernumbers $\tilde{\mathbb{R}}$, the complex c -numbers $\tilde{\mathbb{C}}_c$, or the real c -numbers $\tilde{\mathbb{R}}_c$, respectively. In each of these sub-algebras, an element is again invertible if and only if its body is invertible, and the inverse is again given by (10).

Supernumbers are not strictly a field because we cannot divide by a supernumber with a zero body. Therefore, when we define algebras over supernumbers we are going beyond the realm of linear algebra to super-commutative algebra. However, supernumbers are still better than rings (like integers) as their body is a field (like real numbers and complex numbers) in the conventional sense and division is well-defined for a non-zero body. So in some sense, supernumbers have a minimum deviation from the notion of a field and we will call them superfield $\tilde{\mathbb{F}}$.

We want to define our superalgebra A over a superfield $\tilde{\mathbb{F}}$. We will demand that the multiplication of any element in the algebra with an a -type supernumber will flip its grading while a c -type supernumber will not change the grading. This can be expressed as:

$$|\alpha x| = |\alpha| + |x| \pmod{2} \quad \forall \alpha \in \tilde{\mathbb{F}}, x \in A \quad (14)$$

Consider a superalgebra A over a *superfield* $\tilde{\mathbb{F}}$ equipped with a bilinear, non-degenerate, super-symmetric inner product $\langle _ | _ \rangle$. Here bilinear means

$$\langle \alpha x + \beta y | z \rangle = \alpha \langle x | z \rangle + \beta \langle y | z \rangle \quad (15a)$$

$$\langle z | \alpha x + \beta y \rangle = \langle z | x \rangle \alpha + \langle z | y \rangle \beta, \quad (15b)$$

non-degenerate means

$$\text{if } \langle x | z \rangle = \langle y | z \rangle \text{ for all } z, \text{ then } x = y, \quad (16)$$

and supersymmetric means

$$\langle x | y \rangle = (-1)^{xy} \langle y | x \rangle \quad (17)$$

for any elements $x, y, z \in A$ and any scalars $\alpha, \beta \in \tilde{\mathbb{F}}$. If we define the quadratic norm

$$N(x) \equiv \langle x | x \rangle, \quad (18)$$

we find

$$[1 + (-1)^{xy}] \langle x | y \rangle = N(x + y) - N(x) - N(y). \quad (19)$$

So note that, in contrast to the usual case, where this equation could be used to define the inner product from the norm, in the super case, it cannot: it only defines $\langle x|y \rangle$ when x and y are not *both* fermions.

Definition: We define a *super-composition algebra* A as a superalgebra that is equipped with a super-symmetric inner-product $\langle _ | _ \rangle$ and satisfies the following:

$$\langle uv|wx \rangle = 2(-1)^{vw} \langle u|w \rangle \langle v|x \rangle - (-1)^{vw+vx+wx} \langle ux|wv \rangle. \quad (20)$$

This is obtained by inserting appropriate minus signs into the exchange law for composition algebras [5].

We are interested in unital super-composition algebras which means we demand there is an identity e_0 in the bosonic subalgebra such that:

$$e_0 x = x e_0 = x \quad \forall x \in A. \quad (21)$$

It follows immediately from plugging $u = v = w = x = e_0$ in equation (20) that

$$\langle e_0|e_0 \rangle = 1. \quad (22)$$

We define the conjugate of x as the element obtained by taking reflection along the identity:

$$x^* = 2\langle x|e_0 \rangle e_0 - x. \quad (23)$$

It is easy to check that this is an involution on the algebra. $*$: $A \rightarrow A$ such that

$$(x^*)^* = x. \quad (24)$$

Furthermore, if we demand that it act as a *super*anti-automorphism $(xy)^* = (-1)^{xy} y^* x^*$ then note that the following expression is always along e_0 :

$$\frac{1}{2}(x^* y + (-1)^{xy} y^* x). \quad (25)$$

If we identify identity with the number 1 in the superfield, the expression above naturally gives a bilinear super-symmetric inner-product:

$$\langle x|y \rangle = \frac{1}{2}(x^* y + (-1)^{xy} y^* x) \in \widetilde{\mathbb{F}}. \quad (26)$$

3 Super-triality and Super-alternativity

The super-composition law implies the existence of a three-form $\langle _ , _ , _ \rangle : A \times A \times A \rightarrow \widetilde{\mathbb{F}}$ given by $\langle x, y, z \rangle = \langle xy, z \rangle$ that satisfies the following supersymmetry:

$$\langle x, y, z \rangle = (-1)^{yz} \langle x^*, z, y \rangle \quad (27a)$$

$$= (-1)^{xy} \langle y^*, x^*, z^* \rangle. \quad (27b)$$

Proof. Let us consider the super-composition law given in equation (20),

$$\langle uv|wx\rangle = 2(-1)^{vw}\langle u|w\rangle\langle v|x\rangle - (-1)^{vw+vx+wx}\langle ux|wv\rangle \quad (28a)$$

$$\Rightarrow \langle uv|x\rangle = 2\langle u|e_0\rangle\langle v|x\rangle - (-1)^{vx}\langle ux|v\rangle \quad (28b)$$

$$\Rightarrow \langle uv|x\rangle = (-1)^{vx}(2\langle u|e_0\rangle\langle x|v\rangle - \langle ux|v\rangle) \quad (28c)$$

$$\Rightarrow \langle uv|x\rangle = (-1)^{vx}\langle u^*x|v\rangle. \quad (28d)$$

Therefore, we have established:

$$\langle u, v, x\rangle = (-1)^{vx}\langle u^*, x, v\rangle. \quad (29)$$

Similarly, we can obtain all the symmetries of the “super-triality” three-form. If we feed only bosonic entries to the three-form it acts as the usual triality three-form for the composition algebras. Further, if the composition algebra is a division algebra (i.e. take the metric to be anisotropic), the three-form becomes the triality of the algebra, defined as a map that reduces to a duality upon fixing one of the entries, as explained in [7].

Let us consider a *supersubgroup* of linear transformations of our superalgebra denoted as $End(A)$ such that this subgroup also preserves the norm of our superalgebra. We use the notation of our particle physics friends of attaching an extra “s” before things to denote the super-version and call the subgroup $SISO(A)$ or the super-isometry group of our superalgebra.

Proposition: If we have a triple of linear invertible maps (g_1, g_2, g_3) such that $g_i \in SISO(A) : A \rightarrow A$, satisfying:

$$\langle x, y, z\rangle = \langle g_1(x), (-1)^{g_2x}g_2(y), (-1)^{g_3(x+y)}g_3(z)\rangle, \quad (30)$$

then the triple belongs to the super-triality group of the super-composition algebra following the generalized super-automorphism rule:

$$g_1(xy) = (-1)^{g_3x}g_2(x)g_3(y), \quad (31)$$

and there exists a triple of generalized super-derivations $(\delta_1, \delta_2, \delta_3)$ in the corresponding Lie superalgebra (called the super-triality algebra), that satisfy:

$$\delta_1(xy) = \delta_2(x)y + (-1)^{\delta_3x}x\delta_3(y). \quad (32)$$

Proof. If the triple (g_1, g_2, g_3) leaves the super-triality three-form invariant then it is enough to show that the first-order transformation caused by the super-triality derivation $(\delta_1, \delta_2, \delta_3)$ is zero.

$$0 = \langle \delta_1(x)y, z\rangle + (-1)^{\delta_2x}\langle x\delta_2(y), z\rangle + (-1)^{\delta_3(x+y)}\langle xy, \delta_3(z)\rangle \quad (33a)$$

$$= \langle \delta_1(x)y, z\rangle + (-1)^{\delta_2x}\langle x\delta_2(y), z\rangle - \langle \delta_3(xy), z\rangle \quad (33b)$$

$$\Rightarrow \langle \delta_3(xy), z\rangle = \langle \delta_1(x)y, z\rangle + (-1)^{\delta_2x}\langle x\delta_2(y), z\rangle, \quad \forall z \in A \quad (33c)$$

$$\Rightarrow \delta_3(xy) = \delta_1(x)y + (-1)^{\delta_2x}x\delta_2(y). \quad (33d)$$

In this proof, we have assumed that each δ_i preserves the inner-product to get to the second line ($\delta_i \in \mathfrak{siso}(A)$). The non-degeneracy of the inner-product is needed to get

to the last line of the proof. Therefore, we note that the triplet $(\delta_1, \delta_2, \delta_3)$ forms the super-triality derivation.

The super-automorphism group is a subgroup of the super-triality group corresponding to the case when the triple (g_1, g_2, g_3) are all the same (g, g, g) and:

$$g(xy) = (-1)^{g_x} g(x)g(y). \quad (34)$$

The corresponding Lie superalgebra is the super-derivation algebra generated by a δ satisfying:

$$\delta(xy) = \delta(x)y + (-1)^{\delta_x} x\delta(y). \quad (35)$$

$\mathbb{R}, \mathbb{C}, \mathbb{H}$, and their split versions are associative but the octonions and split-octonions are not. However, all composition algebras satisfy Moufang's law which implies power-associativity. Artin's theorem also shows that all composition algebras are alternative [8]. An alternative algebra is one where the associator given below is completely anti-symmetric in all its arguments:

$$[x, y, z] := (xy)z - x(yz) = -[y, x, z] = -[x, z, y]. \quad (36)$$

We will demand our super-composition algebra to be super-alternative such that the associator defined same as above is *super*anti-symmetric in all its arguments:

$$[x, y, z] = (-1)^{xy} [y, x, z] = (-1)^{yz} [x, z, y]. \quad (37)$$

4 Examples

We tried to construct a super-composition algebra over the field of real numbers and complex numbers. During the construction, we realized that if we wanted the identity to be orthogonal to other vectors the metric must be symmetric in the bosonic part, skew-symmetric in the fermionic part, and trivial for a boson-fermion pair. Okubo showed in his 1983 paper that the only super-composition algebras over a characteristic 0 (like the reals) are the homogenous bosonic composition algebras [9]. Further, Elduque and Okubo in 2002 found super-composition algebras over fields of non-zero characteristic [10].

However, over the superfield $\tilde{\mathbb{R}}$ or $\tilde{\mathbb{C}}$ it is possible to define super-composition algebras that are not purely bosonic. In fact, the only non-homogenous super-composition algebras with a finite basis are the four normed division algebras $\tilde{\mathbb{R}}_{\mathbb{C}}, \tilde{\mathbb{C}}_{\mathbb{C}}, \tilde{\mathbb{H}}_{\mathbb{C}}$, and $\tilde{\mathbb{O}}_{\mathbb{C}}$ defined over the superfield of supernumbers $\tilde{\mathbb{C}}$. And the seven real forms of these $\tilde{\mathbb{R}}, \tilde{\mathbb{C}}, \tilde{\mathbb{H}}, \tilde{\mathbb{O}}$, and their split forms $\tilde{\mathbb{C}}', \tilde{\mathbb{H}}'$, and $\tilde{\mathbb{O}}'$ defined over the superfield of $\tilde{\mathbb{R}}$.

To prove this we will use the result in [9], which states that the only super-composition algebras over a characteristic zero field are the purely bosonic composition algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$, and their split forms.

Let us take a finite (p, q) dimensional super-composition algebra spanned by the basis

$$e_A = \{e_{\alpha 1}, \dots, e_{\alpha p}, e_{a 1}, \dots, e_{a q}\}$$

where the $e_{\alpha i}$ s span the bosonic subspace whereas $e_{a i}$ s span the fermionic subspace. Without loss of generality, we can take four elements of the superalgebra (let's say u, v, w, x)

such that their superfield valued coefficients have a zero soul (which is the same as working with a characteristic 0 field). In that case, equation(20) will put constraints on the algebra product, essentially demanding that the algebra is purely bosonic [9]. Therefore, there are no fermionic basis vectors, *i.e.*, $q = 0$ and the only allowed values of p are 1, 2, 4, and 8 corresponding to the composition algebras.

Now, consider the real super-composition algebra $\tilde{\mathbb{K}}$ defined to be the composition algebra $\mathbb{K}$ over the supernumbers $\tilde{\mathbb{F}} = (\tilde{\mathbb{R}} \text{ or } \tilde{\mathbb{C}})$. We can write an element $x \in \tilde{\mathbb{K}}$ as

$$x = x^0 e_0 + x^i e_i = x^\alpha e_\alpha = (x_c^\alpha + x_a^\alpha) e_\alpha \quad (x^\alpha \in \tilde{\mathbb{F}}, x_c^\alpha \in \tilde{\mathbb{F}}_c, x_a^\alpha \in \tilde{\mathbb{F}}_a). \quad (38)$$

Here e_α are the basis vectors for the algebra $\mathbb{K}$ with e_0 being the identity. The part with only c-type numbers as coefficients forms the bosonic subalgebra whereas the part with only a-type numbers as coefficients forms the fermionic subalgebra.

Let $f_{\beta\gamma}^\alpha$ be the multiplication structure constants for the composition algebra such that:

$$xy = (xy)^\alpha e_\alpha = (x^\beta e_\beta)(y^\gamma e_\gamma) = x^\beta y^\gamma f_{\beta\gamma}^\alpha e_\alpha. \quad (39)$$

Working in this basis with the given structure constants, the super-composition law in equation(20) reduces to:

$$\begin{aligned} \langle uv|wx \rangle &= 2(-1)^{vw} \langle u|w \rangle \langle v|x \rangle - (-1)^{vw+vx+wx} \langle ux|wv \rangle \\ \langle u^\alpha v^\beta f_{\alpha\beta}^\sigma e_\sigma | w^\gamma x^\delta f_{\gamma\delta}^\rho e_\rho \rangle &= 2(-1)^{wv} \langle u^\alpha e_\alpha | w^\gamma e_\gamma \rangle \langle v^\beta e_\beta | x^\delta e_\delta \rangle - (-1)^{xw+xv+vw} \langle u^\alpha x^\delta f_{\alpha\delta}^\sigma e_\sigma | w^\gamma v^\beta f_{\gamma\beta}^\rho e_\rho \rangle \\ \Rightarrow u^\alpha v^\beta w^\gamma x^\delta (f_{\alpha\beta}^\sigma f_{\gamma\delta}^\rho g_{\sigma\rho}) &= (-1)^{vw} u^\alpha w^\gamma v^\beta x^\delta (2g_{\alpha\gamma} g_{\beta\delta}) - (-1)^{wv+wx+vx} u^\alpha x^\delta w^\gamma v^\beta (f_{\alpha\delta}^\sigma f_{\gamma\beta}^\rho g_{\sigma\rho}) \\ \Rightarrow u^\alpha v^\beta w^\gamma x^\delta (f_{\alpha\beta}^\sigma f_{\gamma\delta}^\rho g_{\sigma\rho}) &= u^\alpha v^\beta w^\gamma x^\delta (2g_{\alpha\gamma} g_{\beta\delta} - f_{\alpha\delta}^\sigma f_{\gamma\beta}^\rho g_{\sigma\rho}) \end{aligned} \quad (40)$$

Here, the last line follows from the supernumbers being super-commutative $u^\alpha v^\beta = (-1)^{|u||v|} v^\beta u^\alpha$. The last line is identically true for all values of $u^\alpha, v^\beta, w^\gamma, x^\delta$ since the structure constants of a composition algebra satisfy the relation $f_{\alpha\beta}^\sigma f_{\gamma\delta}^\rho g_{\sigma\rho} = 2g_{\alpha\gamma} g_{\beta\delta} - f_{\alpha\delta}^\sigma f_{\gamma\beta}^\rho g_{\sigma\rho}$. Therefore, the super-composition algebras are restricted to be $\tilde{\mathbb{R}}, \tilde{\mathbb{C}}, \tilde{\mathbb{H}}, \tilde{\mathbb{O}}, \tilde{\mathbb{C}}', \tilde{\mathbb{H}}', \tilde{\mathbb{O}}'$, and their complexifications.

Let us recall the conjugation operation:

$$x^* = 2\langle x|e_0 \rangle e_0 - x = x^0 e_0 - x^i e_i, \quad (41)$$

$$(xy)^* = (-1)^{xy} y^* x^*. \quad (42)$$

The inner product on this super-composition algebra $g : \tilde{\mathbb{K}} \times \tilde{\mathbb{K}} \rightarrow \tilde{\mathbb{R}}$ in terms of the conjugation operation is:

$$g(x, y) = \langle x, y \rangle = \frac{1}{2} [x^* y + (-1)^{xy} y^* x] = (x^\alpha g_{\alpha\beta} y^\beta) e_0 \cong x^\alpha y_\alpha = x_\alpha y^\alpha. \quad (43)$$

Here $g_{\alpha\beta}$ is the super-symmetric bilinear metric defined for the underlying algebra $\mathbb{K}$. It is positive-definite for the division algebras but has a split-signature for the split-composition algebras.

Lie superalgebras over a characteristic zero field were first discussed by Kac [4]. Here we consider a more general version of Lie superalgebras over supernumbers. A Lie superalgebra is a superalgebra equipped with a super-*antisymmetric* product that acts as

a superderivation:

$$ab = -(-1)^{ab}ba \quad (44a)$$

$$a(bc) = (ab)c + (-1)^{ab}b(ac). \quad (44b)$$

The super-derivation rule is also called the super-Jacobi identity.

The Lie superalgebras that preserve the super-symmetric inner-product (43) for the super-composition algebras $\tilde{\mathbb{K}}$ are given by $\mathfrak{g}_{\tilde{\mathbb{K}}}$, where $\mathfrak{g}_{\mathbb{K}}$ is the Lie algebra that preserves the symmetric inner-product for the composition algebra $\mathbb{K}$ (known to be $\mathfrak{so}(\mathbb{K})$ for composition algebras) and $\mathfrak{g}_{\tilde{\mathbb{K}}}$ is $\mathfrak{g}_{\mathbb{K}}$ valued over supernumbers.

Let the Lie superalgebra $\mathfrak{g}_{\tilde{\mathbb{K}}}$ be spanned by the generators $t_{\alpha\beta}$. The infinitesimal transformation caused by the generators are $\delta x^\alpha = t^\alpha_{\beta} x^\beta$ or $\delta x_\alpha = t_{\alpha\beta} x^\beta$. If this transformation preserves the inner-product then we have:

$$\begin{aligned} 0 = \delta g(x, y) &= \delta(x_\alpha y^\alpha) \\ &= (\delta x_\alpha) y^\alpha + (-1)^{x\delta} x^\alpha (\delta y_\alpha) \\ &= t_{\beta\alpha} x^\alpha y^\beta + (-1)^{xt} x^\alpha t_{\alpha\beta} y^\beta \end{aligned}$$

or, in other words,

$$t_{\alpha\beta} = -t_{\beta\alpha}, \quad (45)$$

i.e. $t_{\alpha\beta}^c = -t_{\beta\alpha}^c$ and $t_{\alpha\beta}^a = -t_{\beta\alpha}^a$, corresponding to the Lie superalgebra $\mathfrak{so}(\tilde{\mathbb{K}}) \cong \mathfrak{so}(\mathbb{K}; \tilde{\mathbb{R}})$. (We use Baez's [7] notation, where $\mathfrak{so}(\mathbb{K})$ is $\mathfrak{so}(2)$ for $\mathbb{C}$, $\mathfrak{so}(4)$ for $\mathbb{H}$, and $\mathfrak{so}(8)$ for $\mathbb{O}$)

Recall the fermionic conjugation given in equation (12b). It acts in the following way on the super-composition algebras:

$$x' = (x_c^\alpha - x_a^\alpha) e_\alpha, \quad (46)$$

$$(xy)' = x'y'. \quad (47)$$

Another interesting inner-product to consider is the inner-product on the super-composition algebra $\tilde{\mathbb{K}}$ to the c-type numbers $\tilde{\mathbb{R}}_c$. $\tilde{g} : \tilde{\mathbb{K}} \times \tilde{\mathbb{K}} \rightarrow \tilde{\mathbb{R}}_c$:

$$\tilde{g}(x, y) = \frac{1}{2}[g(x, y) + g(x, y)'] = x_c^\alpha g_{\alpha\beta} y_c^\beta + x_a^\alpha g_{\alpha\beta} y_a^\beta. \quad (48)$$

This inner-product is symmetric in the bosonic part, skew-symmetric in the fermionic part, and is zero for a mixed bosonic and fermionic entry. This inner-product will not satisfy the super-composition rule but is still interesting to consider since it lets us treat the n -dimensional superalgebra $\tilde{\mathbb{K}}$ over $\tilde{\mathbb{R}}$ as a $2n$ -dimensional superalgebra $(\tilde{\mathbb{K}}_{2n})$ over $\tilde{\mathbb{R}}_c$.

Let $t_{\alpha\beta}^1$ be a generator of the Lie-superalgebra that transforms the bosons into bosons, t_{ab}^2 is a generator that transforms fermions to fermions, whereas $t_{a\beta}^3$ and t_{ab}^4 mix the two spaces. We note that $t_{\alpha\beta}^1$ and t_{ab}^2 are in the bosonic subalgebra and are valued in $\tilde{\mathbb{R}}_c$, whereas $t_{a\beta}^3$ and t_{ab}^4 are in the fermionic subalgebra and are valued in $\tilde{\mathbb{R}}_a$. We can write all these generators in a matrix and define its super-transpose:

$$t_{AB} := \begin{pmatrix} t_{\alpha\beta}^1 & t_{ab}^4 \\ t_{a\beta}^3 & t_{ab}^2 \end{pmatrix} \quad (t^T)_{BA} := \begin{pmatrix} t_{\beta\alpha}^2 & t_{\beta a}^3 \\ -t_{b\alpha}^4 & t_{ba}^1 \end{pmatrix}. \quad (49)$$

The infinitesimal transformations generated by this generator can be expressed as $\delta x_A = t_{AB}x^B = x^B t_{BA}^T$.

If the inner-product given in equation (48) is invariant under these transformations, we have:

$$0 = \delta \tilde{g}(x, y) = \frac{1}{2}[\delta x_A y^A + (-1)^{x\delta} x^A \delta y_A + (\delta x'_A) y^{A'} + (-1)^{x\delta} x'^A \delta y'_A] \quad (50)$$

or, in other words

$$t_{\alpha\beta}^1 = -t_{\beta\alpha}^1 \quad (51a)$$

$$t_{ab}^2 = -t_{ba}^2 \quad (51b)$$

$$t_{a\beta}^3 = -t_{b\alpha}^4. \quad (51c)$$

This corresponds to the Lie superalgebra with a bosonic part $\mathfrak{so}(\mathbb{K}; \tilde{\mathbb{R}}_c) \oplus \mathfrak{so}(\mathbb{K}; \tilde{\mathbb{R}}_c)$, and an n^2 -dimensional fermionic part valued over $\tilde{\mathbb{R}}_a$. We denote this Lie superalgebra as $\mathfrak{so}(\mathbb{K}|\mathbb{K})$.

How do we know that $\mathfrak{so}(\tilde{\mathbb{K}})$ and $\mathfrak{so}(\mathbb{K}|\mathbb{K})$ are Lie-superalgebras and satisfy equation (44)? For $\mathfrak{so}(\tilde{\mathbb{K}})$, it is easy to see this since the basis for this superalgebra are homogenous bosonic generators for the $\mathfrak{so}(\mathbb{K})$ Lie-algebra and the supernumber valued coefficients take care of the grading to make it a superalgebra.

The Lie-superalgebra $\mathfrak{so}(\mathbb{K}|\mathbb{K})$ can be understood by selectively grading the Lie-algebra $\mathfrak{so}(2\mathbb{K})$. It is well known that $\mathfrak{so}(2n) = \mathfrak{so}(n) \oplus \mathfrak{so}(n) \oplus \mathbb{R}^n \otimes \mathbb{R}^n$. If we take the coefficients of $\mathfrak{so}(n) \oplus \mathfrak{so}(n)$ subalgebra to be c-type numbers and the coefficients of $\mathbb{R}^n \otimes \mathbb{R}^n$ subalgebra to be a-type coefficients then they form a Lie-superalgebra with the appropriate grading coming from the supernumbers.

$$\mathfrak{so}(n|n) = \begin{pmatrix} \mathfrak{so}(n) & 0 \\ 0 & \mathfrak{so}(n) \end{pmatrix} \oplus \begin{pmatrix} 0 & a \in \tilde{\mathbb{R}}_c^n \otimes \tilde{\mathbb{R}}_a^{n*} \\ -a \in \tilde{\mathbb{R}}_a^n \otimes \tilde{\mathbb{R}}_c^{n*} & 0 \end{pmatrix} \quad (52)$$

Here $\tilde{\mathbb{R}}_c^{n*}$ is the dual of $\tilde{\mathbb{R}}_c^n$ that maps it to a c-type number. Note that $\tilde{\mathbb{R}}_c^{n*}$ is isomorphic to $\tilde{\mathbb{R}}_c^n$ but we are making the distinction to be more pedantic. More technically we could write the off-diagonal elements as $Hom(\tilde{\mathbb{R}}_c^n, \tilde{\mathbb{R}}_c^n)$.

5 Superoctonions and the (Euclidean) Standard Model

This section will particularly focus on superoctonions $\tilde{\mathbb{O}}$. $\mathfrak{so}(8, \tilde{\mathbb{R}})$ is the Lie-superalgebra that preserves the norm of superoctonions and is also their supertriality algebra. The Lie-superalgebra that preserves the norm from $\tilde{\mathbb{O}}$ to $\tilde{\mathbb{R}}_c$ is $\mathfrak{so}(8|8)$.

Previous work in non-commutative geometry [11, 12] shows that the triple $\mathcal{A} = \{q, m, z\}$ defined at each point of spacetime $\mathcal{M}$ along with the Hilbert space $\mathcal{H}$ of square-integrable Dirac spinors on $\mathcal{M}$ is sufficient to describe the Standard Model fermions and bosons. Here $q \in \mathbb{H}$ is a quaternions, $m \in \mathbb{C}[3]$ is a 3×3 complex matrix and $z \in \mathbb{C}$ is a complex number.

An explanation of the Standard Model coming from the underlying mathematical structure of spectral triples is good but it has to be artificially imposed. We will now explain

how the Standard Model with local Lorentz symmetry can be obtained from a superoctonionic bundle over spacetime. Choosing a complex structure and a quaternionic structure in this bundle will naturally create the triple $\mathcal{A}$ and the Hilbert space $\mathcal{H}$ needed in non-commutative geometry.

We work with a $\mathcal{C}^\infty$ differentiable Hausdorff manifold $\mathcal{M}$ that locally looks like $\mathbb{R}^4$. At each point x_p of this manifold, we define a superoctonionic module $\tilde{\mathcal{O}}_p$ and its endomorphisms $End(\tilde{\mathcal{O}}_p)$. Note that this is similar to choosing a *spin*-structure $\mathcal{S}_{2n}$ on a manifold $\mathcal{M}$, where the linear transformations of the spinor form the Clifford algebra $Cl(2n) \cong End(\mathcal{S}_{2n})$.

Another important point to note is that we are not choosing our manifold to be a supermanifold as described in [5, 6]. A supermanifold looks locally like $\mathbb{R}^{n|m}$ and the tangent plane is described by n vectors and m Grassmann odd variables. In a flat supermanifold the spacetime Poincaré symmetry is extended with the super-Poincaré symmetry $ISO(n|m)$ that includes supercharges. We are just choosing the fiber $\tilde{\mathcal{O}}_p$ in our bundle to be $\mathbb{Z}_2$ graded but the manifold itself is not and the local spacetime symmetry of the tangent space remains to be $SO(4)$.

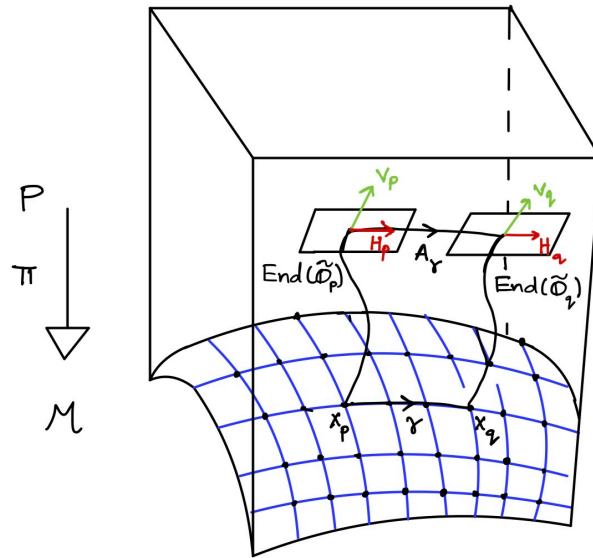


Figure 1: The superoctonionic bundle P looks locally like $\mathcal{M} \times End(\tilde{\mathcal{O}})$. The sub-bundle with vectors V_i along the direction of the fibers is called the vertical bundle whereas the complementary sub-bundle with vectors H_i perpendicular to the fiber is called the horizontal bundle or Ehrrassmann bundle. A_γ is an Ehrrassmann connection and helps in parallel transport from $End(\tilde{\mathcal{O}}_p)$ to $End(\tilde{\mathcal{O}}_q)$. We will see later that the gauge fields live in the Ehrrassmann bundle of P .

5.1 Representation

Let us for the moment, focus on just one point of spacetime and look at the algebra of superoctonions. Going from one point to another would require us to use exterior

derivatives and Ehrassmann connections. The norm from $\tilde{\mathcal{O}}$ to $\tilde{\mathbb{R}}_c$ is given by:

$$\langle x|y \rangle = x_c^\alpha g_{\alpha\beta} y_c^\beta + x_a^\alpha g_{\alpha\beta} y_a^\beta. \quad (53)$$

We have seen from a more general case before that the Lie superalgebra that preserves this norm is $\mathfrak{so}(8|8)$ which can be decomposed in the bosonic and fermionic subalgebra as follows:

$$\mathfrak{so}(8|8) = \begin{pmatrix} \mathfrak{so}(8) & 0 \\ 0 & \mathfrak{so}(8) \end{pmatrix} \oplus \begin{pmatrix} 0 & a \in \tilde{\mathbb{R}}_a^8 \otimes \tilde{\mathbb{R}}_c^8 \\ -a \in \tilde{\mathbb{R}}_a^8 \otimes \tilde{\mathbb{R}}_c^8 & 0 \end{pmatrix} \quad (54)$$

The Lie-superalgebra $\mathfrak{so}(8|8)$ is a subalgebra of $End(\tilde{\mathcal{O}})$ and it's action on the superoc-tonionic module can depicted as follows:

$$\begin{pmatrix} \tilde{\mathcal{O}}'_c \\ \tilde{\mathcal{O}}'_a \end{pmatrix} = \begin{pmatrix} a \in \mathfrak{so}(8) & b \in \tilde{\mathcal{O}}_c \otimes \tilde{\mathcal{O}}_a^* \\ -b \in \tilde{\mathcal{O}}_a \otimes \tilde{\mathcal{O}}_c^* & c \in \mathfrak{so}(8) \end{pmatrix} \begin{pmatrix} \tilde{\mathcal{O}}_c \\ \tilde{\mathcal{O}}_a \end{pmatrix} \quad (55)$$

Now we want to choose a complex structure $J \in End(\tilde{\mathcal{O}}_a)$ such that $J^2 = -1$. Choosing this complex structure will reduce the space $\tilde{\mathcal{O}}_a$ to $\tilde{\mathcal{C}}_a^4$. The subalgebra of $\mathfrak{so}(8)$ that commutes with J and hence leaves it invariant is $\mathfrak{su}(4)$.

Similarly, we want to re-write the bosonic part $\tilde{\mathcal{O}}_c$ as a pair of c-type quaternions $\tilde{\mathbb{H}}_c \oplus \tilde{\mathbb{H}}_c$ which we know is possible form the Cayley-Dickson process. The subalgebra of $\mathfrak{so}(8)$ that respects this division and does not mix the two sub-space is $\mathfrak{so}(4) \oplus \mathfrak{so}(4)$. Note that $\mathfrak{so}(4) \oplus \mathfrak{so}(4)$ commutes with the chirality operator $\chi = diag(1, 1, 1, 1, -1, -1, -1, -1)$.

After the making the above choice, our supermodule breaks into $\tilde{\mathbb{H}}_c \oplus \tilde{\mathbb{H}}_c$ in the bosonic part and $\tilde{\mathcal{C}}_a^4$ in the fermionic part. Let us call the *supersubalgebra* of $\mathfrak{so}(8|8)$ that respects these algebraic structures as $\mathcal{A}$. The action of $\mathcal{A}$ on the supermodule can be depicted as:

$$\begin{pmatrix} \tilde{\mathbb{H}}'_c \oplus \tilde{\mathbb{H}}'_c \\ \tilde{\mathcal{C}}'^4_a \end{pmatrix} = \begin{pmatrix} a \in \mathfrak{so}(4) \oplus \mathfrak{so}(4) & b \in (\tilde{\mathbb{H}}_c \oplus \tilde{\mathbb{H}}_c) \otimes \tilde{\mathcal{C}}_a^{4*} \\ -b^* \in \tilde{\mathcal{C}}_a^4 \otimes (\tilde{\mathbb{H}}_c \oplus \tilde{\mathbb{H}}_c)^* & c \in \mathfrak{su}(4) \end{pmatrix} \begin{pmatrix} \tilde{\mathbb{H}}_c \oplus \tilde{\mathbb{H}}_c \\ \tilde{\mathcal{C}}_a^4 \end{pmatrix} \quad (56)$$

It can be proved that this superalgebra is closed. The super-algebra $\mathcal{A}$ contains all the Standard Model gauge bosons that are Lie-algebra valued in the bosonic subalgebra ($\mathcal{A}_0$) and the Grassmann-valued Standard Model fermions transforming as the fundamental representation of these Lie-algebras live in the fermionic subalgebra ($\mathcal{A}_1$).

One of the $\mathfrak{so}(4)$ is the local Euclidean spacetime rotation and in equation (56) we are depicting its spinor representation. The other $\mathfrak{so}(4)$ is also written in the spinor representation and can be written as $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ gauge symmetry. The $\mathfrak{su}(4)$ gauge symmetry is isomorphic to $\mathfrak{so}(6)$ and can be further broken to give $\mathfrak{su}(3)_C \oplus \mathfrak{u}(1)_{B-L}$. Therefore the bosonic subalgebra gives us a unification of the Pati-Salam gauge group with the Euclidean group of rotations.

$$\mathcal{A}_0 = \mathfrak{so}(4) \oplus \mathfrak{so}(4) \oplus \mathfrak{so}(6). \quad (57)$$

The fermionic subalgebra $\mathcal{A}_1$ contains all the Standard Model fermions with Euclidean spinor index making a total of 32 complex degrees of freedom. Octonions transform as

Majorana-Weyl fermions of $SO(8)$ denoted by 8_s which is equivalent to the vector representation 8_v due to triality of $\mathfrak{so}(8)$. The fermions indeed transform as the fundamental representations of the gauge groups and it can be checked from the branching rules of $\mathfrak{so}(8)$:

$$8 = (1, 2, 1, 2) \oplus (2, 1, 2, 1) \quad \text{branching into } \mathfrak{su}(2) \otimes \mathfrak{su}(2) \otimes \mathfrak{su}(2) \otimes \mathfrak{su}(2) \quad (58)$$

$$8 = 4(1) \oplus \bar{4}(-1) \quad \text{branching into } \mathfrak{su}(4) \otimes \mathfrak{u}(1). \quad (59)$$

Therefore, we see that $\tilde{\mathcal{O}}_c$ breaks into left(right)-handed spinors of spacetime isometry $\mathfrak{so}(4)$ that are also a doublets of $\mathfrak{su}(2)_L$ ($\mathfrak{su}(2)_R$). Simultaneously $\tilde{\mathcal{O}}_a$ breaks into a spinor of $\mathfrak{su}(4)$ which contains the color and $B - L$ symmetry. The fermionic subalgebra that was a tensor product of $\tilde{\mathcal{O}}_c$ and $\tilde{\mathcal{O}}_a$ breaks into the desired fermions of SM. The upper off-diagonal matrix transforms as the fermions while the lower block diagonal matrix transforms as the anti-fermions.

$$\mathcal{A}_1 = \left(\begin{array}{c|cccc} & u_L^A & u_L^A & u_L^A & \nu_L^A \\ & d_L^A & d_L^A & d_L^A & e_L^A \\ & u_R^{A'} & u_R^{A'} & u_R^{A'} & \nu_R^{A'} \\ & d_R^{A'} & d_R^{A'} & d_R^{A'} & e_R^{A'} \\ \hline \bar{u}_L^{A'} & \bar{d}_L^{A'} & \bar{u}_R^A & \bar{d}_R^A & \\ \bar{u}_L^{A'} & \bar{d}_L^{A'} & \bar{u}_R^A & \bar{d}_R^A & \\ \bar{u}_L^{A'} & \bar{d}_L^{A'} & \bar{u}_R^A & \bar{d}_R^A & \\ \bar{\nu}_L^{A'} & \bar{e}_L^{A'} & \bar{\nu}_R^A & \bar{e}_R^A & \end{array} \right) \quad (60)$$

In the above equation, A denotes the left-handed spinor index, whereas, A' denotes the right-handed spinor index. The fermions in the fermionic subalgebra transform as fundamental representations of the Lie groups in the bosonic subalgebra. The charge for the fermions under various gauge groups is listed below.

	$\mathfrak{so}(4)$	$\mathfrak{su}(3)$	$\mathfrak{su}(2)_L$	$\mathfrak{su}(2)_R$	$\mathfrak{u}(1)_{B-L}$
q_L	(2, 0)	3	2	1	1/3
l_L	(2, 0)	1	2	1	-1
q_R	(0, 2)	3	1	2	1/3
l_R	(0, 2)	1	1	2	-1

To get the Standard Model with hypercharge we will use the well-known trick to break $\mathfrak{su}(2)_R$ by choosing one of its generators and calling the subalgebra $\mathfrak{u}(1)_X$. In doing so, the fermions gain the following charge.

	$\mathfrak{so}(4)$	$\mathfrak{su}(3)$	$\mathfrak{su}(2)_L$	$\mathfrak{u}(1)_X$	$\mathfrak{u}(1)_{B-L}$
q_L	(2, 0)	3	2	0	1/3
l_L	(2, 0)	1	2	0	-1
u_R	(0, 2)	3	1	1/2	1/3
d_R	(0, 2)	3	1	-1/2	1/3
ν_R	(0, 2)	1	1	1/2	-1
e_R	(0, 2)	1	1	-1/2	-1

Finally, the hypercharge Y is given by:

$$Y = X + (B - L)/2, \quad (61)$$

so we have obtained the Standard Model:

	$\mathfrak{so}(4)$	$\mathfrak{su}(3)$	$\mathfrak{su}(2)_L$	$\mathfrak{u}(1)_Y$
q_L	$(2, 0)$	3	2	1/6
l_L	$(2, 0)$	1	2	-1/2
u_R	$(0, 2)$	3	1	2/3
d_R	$(0, 2)$	3	1	-1/3
ν_R	$(0, 2)$	1	1	0
e_R	$(0, 2)$	1	1	-1

5.2 Action

We have seen that the superoctonionic bundle over spacetime can be broken systematically to give the Standard Model gauge bosons and fermions living at each point of spacetime in a superalgebra $\mathcal{A}$. As seen in figure(1), the bundle itself can be broken into the vertical bundle and the horizontal bundle (Ehressmann bundle). The gauge bosons live in the horizontal bundle and help in parallel transportation from different spacetime points. Therefore, the gauge bosons are 1-forms of the spacetime co-tangent bundle and form the Ehressmann connections for the different gauge groups. In general, we can write the connection as:

$$D = d + A, \quad (62)$$

for $A_\mu \in \mathcal{A}_0$. A also includes the spin connection ω for the local Euclidean rotation $\mathfrak{so}(4)$ apart from the internal symmetries. The spin connection is just a $\mathfrak{so}(4)$ Lie-algebra valued 1-form that is related to the tetrads through the torsionless relation:

$$de^a + \omega^a_b e^b = 0. \quad (63)$$

The tetrad is also a 1-form that maps vectors in the tangent space of spacetime to the vector space of local $\mathfrak{so}(4)$ transformations.

To construct the action we need some guiding principles. Firstly, the action should be a c-type number, therefore the fermion fields should appear in even powers. Secondly, the action should be integrated over all of spacetime, therefore we need to construct 4-forms from our field content. Lastly, the action should be invariant under gauge transformations. The following is a standard inner-product from the space of p -forms to the field:

$$(p, q) := \int p * q; \quad \forall p, q \in \Omega^p(\mathcal{M}). \quad (64)$$

Therefore, if we denote the fermions as $\psi \in \mathcal{A}_1$ then the most general action ignoring irrelevant operators is given by:

$$S = a(DA, DA) + b(DA, *DA) + c(\bar{\psi}e, \gamma D\psi) + f(\bar{\psi}, \psi). \quad (65)$$

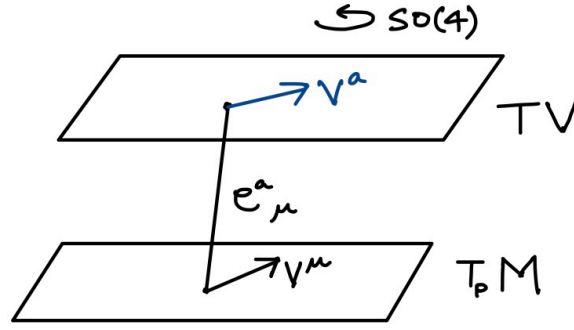


Figure 2: Tetrad e^a_μ maps from the spacetime tangent space to the vector space on which the local $\mathfrak{so}(4)$ acts.

The first term is just the Yang-Mills action, the second term is topological, and the third term is the Dirac action with $\gamma \in \text{Cliff}(4) \cong \text{End}(\mathcal{S}_4)$. The last term corresponds to the mass of the fermions.

To contract the gauge indices we can use the Killing-form for the various gauge groups. For Euclidean rotation $\mathfrak{so}(4)$ we can use the two invariant tensor η_{ab} and ϵ_{abcd} to contract the gauge indices. The action above boils down to the Standard Model action with some changes in the gravity sector that we explicitly discuss in the next subsection.

5.3 Higher-Derivative Gravity

The Riemann two-form in terms of the spin connection is given by:

$$R^{ab} = D\omega^{ab} = d\omega^{ab} + \omega^a_c \wedge \omega^{cb}. \quad (66)$$

Looking at the first two terms in the action [\[65\]](#), there are four possible terms for gravity:

$$\epsilon_{abcd} R^{ab} \wedge R^{cd}, \quad \eta_{ac} \eta_{bd} R^{ab} \wedge *R^{cd}, \quad \eta_{ac} \eta_{bd} R^{ab} \wedge R^{cd}, \quad \epsilon_{abcd} R^{ab} \wedge *R^{cd}. \quad (67)$$

The first term is the topological Gauss-Bonnet term that will not contribute to the equations of motion and will contribute only if there is some non-trivial boundary:

$$\int \epsilon_{abcd} R^{ab} \wedge R^{cd} = \int d^4x \sqrt{g} (R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta} - 4R_{\mu\nu} R^{\mu\nu} + R^2). \quad (68)$$

The second term looks like a Yang-Mills version of gravity. On simplifying, it turns out to be:

$$\int \eta_{ac} \eta_{bd} R^{ab} \wedge *R^{cd} = \int d^4x \sqrt{g} R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta}. \quad (69)$$

The third and the fourth term look like the following:

$$\int \eta_{ac} \eta_{bd} R^{ab} \wedge R^{cd} = \int d^4x R^{\mu\nu}_{\alpha\beta} R_{\mu\nu\gamma\delta} \epsilon^{\alpha\beta\gamma\delta} \quad (70)$$

$$\int \epsilon_{abcd} R^{ab} \wedge *R^{cd} = \int d^4x \sqrt{g} R^{\mu\nu}_{\alpha\beta} R_{\mu\nu\gamma\delta} \epsilon^{\alpha\beta\gamma\delta}. \quad (71)$$

Considering all four terms, the most general higher-derivative action for gravity can be written as:

$$S = \frac{1}{2\beta} \int d^4x \sqrt{g} (C^2 - \frac{2\alpha}{3} R^2 + \gamma E), \quad (72)$$

here, β, α, γ are a dimensionless coupling constants. C^2 is the Weyl squared term $C^{abcd}C_{abcd}$, and E is the Gauss-Bonnet term.

There are good reasons (based on observations from neutron stars) to believe that the Einstein-Hilbert action is the theory of gravity with no higher derivative terms. How then do we justify our action? We get the above action because we are taking seriously the local spacetime symmetry of $\mathfrak{so}(4)$. But this is not strictly the local spacetime symmetry, we live in a universe that has a negligible but positive cosmological constant Λ . The way to think of this is to embed a 4-sphere with a radius that goes as the inverse of Λ in 5-dimensions. In terms of Cartan geometry, this can be expressed as:

$$\mathcal{M}_{dS} = \frac{SO(5)}{SO(4)}. \quad (73)$$

The spin connection with $\mathfrak{so}(5)$ indices that captures this fact is given by:

$$\omega^{AB} = \begin{pmatrix} \omega^{ab} & \sqrt{\Lambda} e^c \\ -\sqrt{\Lambda} e_c & 0 \end{pmatrix}. \quad (74)$$

Here the capital letter indices are the $\mathfrak{so}(5)$ indices whereas the small letter indices are the $\mathfrak{so}(4)$ indices. If the spacetime is torsionless then the Riemann two-form for this space can be calculated to be:

$$\tilde{R}^A{}_B = D\omega^A{}_B = \begin{pmatrix} R^a{}_b - \Lambda e^a \wedge e_b & 0 \\ 0 & 0 \end{pmatrix}. \quad (75)$$

Or in other words $\tilde{R}^a{}_b = R^a{}_b - \Lambda e^a \wedge e_b$. Plugging this correction in the Reimann two-form back to our four possible terms of the action, we will get the following action:

$$S = \frac{1}{2\beta} \int d^4x \sqrt{g} (C^2 - \frac{2\alpha}{3} R^2 + \gamma E + \Lambda^2 (R - 2\Lambda)). \quad (76)$$

We know that our energy scales $\Lambda^2/\beta \sim 1/G_N$. Therefore we have:

$$\beta \sim \Lambda^2/M_{Pl}^2 \sim 10^{-122}. \quad (77)$$

Therefore the higher derivative term has almost negligible contribution at our energy scales and can be observed only at very high energy scales.

To check the running of the couplings, we need to gauge-fix the action first with BRST symmetry. In [13] and references therein, the authors use the exact renormalization group equation to see the running of the couplings and the result is that the coupling β is asymptotically free, whereas α and γ have a fixed point in the UV. The dimension full couplings Λ and G_N have a gauge-dependent coupling but there product β does not. We are further exploring this to see if coupling gravity with the fields can help remove the Landau pole from the hypercharge beta function. Another interesting area is to take

the metric to be conformally flat, which might be the case if the theory is asymptotically safe. In doing so, the higher derivative gravity action changes to higher derivative action for a dimensionless scalar field Φ that causes the Weyl transformation.

$$g_{\mu\nu}(x) = e^{i\Phi(x)}\eta_{\mu\nu}. \quad (78)$$

Apart from introducing higher derivative terms for gravity, our action in equation (65) also predicts a coupling of the spin connection with the matter fields:

$$\bar{\psi}e_a^\mu\gamma^a\omega_\mu\psi. \quad (79)$$

This interaction is anomaly-free in the Lorentzian signature due to the “good” behavior of $\mathfrak{su}(2)$ algebra. The details can be found in [14].

5.4 Three Generations

Killing and Cartan classified simple compact Lie algebras. They classified four types of classical Lie algebras and five exceptional ones. The classical Lie algebras are related to the division algebras $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$:

$$\mathfrak{so}(n) = \{x \in \mathbb{R}[n], \quad x^\dagger = -x, \quad \text{tr}(x) = 0\} \quad (80a)$$

$$\mathfrak{su}(n) = \{x \in \mathbb{C}[n], \quad x^\dagger = -x, \quad \text{tr}(x) = 0\} \quad (80b)$$

$$\mathfrak{sp}(n) = \{x \in \mathbb{H}[n], \quad x^\dagger = -x, \quad \text{tr}(x) = 0\}, \quad (80c)$$

whereas all the exceptional Lie-algebras are related to the octonions. We have discussed already that G_2 is the automorphism group of the octonions. In 1956, Rosenfeld realized that F_4 was the isometry group of $\mathbb{O}\mathbb{P}^2$. He further conjectures that the exceptional Lie groups E_6 , E_7 , and E_8 might be the isometry groups of $(\mathbb{C} \otimes \mathbb{O})\mathbb{P}^2$, $(\mathbb{H} \otimes \mathbb{O})\mathbb{P}^2$, and $(\mathbb{O} \otimes \mathbb{O})\mathbb{P}^2$ respectively. His conjecture is not precisely true because of the subtleties in describing the projective planes $(\mathbb{H} \otimes \mathbb{O})\mathbb{P}^2$, and $(\mathbb{O} \otimes \mathbb{O})\mathbb{P}^2$. However, there definitely seems to be a link between these exceptional Lie groups and the tensor product of division algebras with octonions. In around 1958 Tits [15] and Freudenthal [16] came up with a construction called the magic square of Lie algebras that nicely decomposes certain Lie algebras in terms of sub-algebras corresponding to two different division algebras. Vinberg constructed the magic square we present here [17] in 1966.

	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	$\mathbb{O}$
$\mathbb{R}$	$\mathfrak{so}(3)$	$\mathfrak{su}(3)$	$\mathfrak{sp}(3)$	$\mathfrak{f}_4$
$\mathbb{C}$	$\mathfrak{su}(3)$	$\mathfrak{su}(3) \oplus \mathfrak{su}(3)$	$\mathfrak{su}(6)$	$\mathfrak{e}_6$
$\mathbb{H}$	$\mathfrak{sp}(3)$	$\mathfrak{su}(6)$	$\mathfrak{so}(12)$	$\mathfrak{e}_7$
$\mathbb{O}$	$\mathfrak{f}_4$	$\mathfrak{e}_6$	$\mathfrak{e}_7$	$\mathfrak{e}_8$

Figure 3: The Magic Square of Lie Algebras.

The Lie algebras in the magic square can be expressed as the triality algebras of the pair of division algebras they correspond to. Frank Adams first presented this in his

lectures and was first documented by Ramond [18]. For a division algebra $\mathbb{K}$ and $\mathbb{K}'$, the corresponding Lie algebra $L(\mathbb{K}, \mathbb{K}')$ from the magic square can be expressed as:

$$L(\mathbb{K}, \mathbb{K}') = \text{Tri}(\mathbb{K}) \oplus \text{Tri}(\mathbb{K}') \oplus \mathbb{K} \otimes \mathbb{K}' \oplus \mathbb{K} \otimes \mathbb{K}' \oplus \mathbb{K} \otimes \mathbb{K}'. \quad (81)$$

In [19], $\mathfrak{e}_6$ was discussed in detail using this formulation. In the triality formulation, $\mathfrak{e}_6$ can be written as $\mathfrak{so}(10) \oplus \mathfrak{u}(1) \oplus (\mathbb{C} \otimes \mathbb{O})^2$. $\mathbb{C} \otimes \mathbb{O}$ transforms as the spinor representation 16 of $\mathfrak{so}(10)$ and notably there are three ways to choose one of the $\mathbb{C} \otimes \mathbb{O}$ to form $\mathfrak{so}(10)$. We are interested in both internal gauge symmetries and the local spacetime symmetry which takes us to $\mathfrak{e}_8$:

$$\mathfrak{e}_8 = \text{Tri}(\mathbb{O}) \oplus \text{Tri}(\mathbb{O}) \oplus \mathbb{O} \otimes \mathbb{O} \oplus \mathbb{O} \otimes \mathbb{O} \oplus \mathbb{O} \otimes \mathbb{O} \quad (82)$$

$$= \mathfrak{so}(8) \oplus \mathfrak{so}(8) \oplus \mathbb{O} \otimes \mathbb{O} \oplus (\mathbb{O} \otimes \mathbb{O})^2 \quad (83)$$

$$= \mathfrak{so}(16) \oplus (\mathbb{O} \otimes \mathbb{O})^2. \quad (84)$$

Note that there is a S_3 symmetry here in choosing one out of the three $\mathbb{O} \otimes \mathbb{O}$ to get $\mathfrak{so}(16)$. This S_3 symmetry possibly corresponds to the S_3 symmetry in the triality algebra discussed in section 3 and can be manifestly seen from the S_3 symmetry in the Dynkin diagram of $SO(8)$.

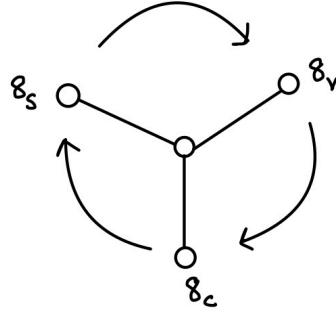


Figure 4: S_3 symmetry in the Dynkin diagram of $SO(8)$.

We have seen already that the $\mathfrak{so}(8|8)$ can be obtained by partially grassmanifying the off-diagonal elements of $\mathfrak{so}(16)$. The superalgebra that we are interested in to get three generations along with $\mathfrak{so}(8|8)$ unification is a subalgebra of $\mathfrak{e}_8(\mathbb{R})$ given by:

$$\mathfrak{e}'_8 = \text{Tri}(\mathbb{O}_c) \oplus \text{Tri}(\mathbb{O}_c) \oplus \mathbb{O}_a \otimes \mathbb{O}_c \oplus \mathbb{O}_a \otimes \mathbb{O}_c \oplus \mathbb{O}_a \otimes \mathbb{O}_c \quad (85)$$

$$= \mathfrak{so}(8) \oplus \mathfrak{so}(8) \oplus \mathbb{O}_a \otimes \mathbb{O}_c \oplus (\mathbb{O}_a \otimes \mathbb{O}_c)^2 \quad (86)$$

$$= \mathfrak{so}(8|8) \oplus (\mathbb{O}_a \otimes \mathbb{O}_c)^2. \quad (87)$$

Thus the superalgebra $\mathfrak{e}'_8$ has the the super unification of the Standard Model bosons and fermions with three ways of obtaining the fermionic content due to the S_3 symmetry.

6 Twistors and Split-Octonions

6.1 Introduction

There seems to be a relationship between twistor space and split-octonions. This was first discovered by Sparling who constructed the Jordan algebra of 3×3 hermitian matrices with split octonionic entries starting from 3 copies of twistor space [20]. More recently, Penrose has tried to construct the algebra of split octonions from quantized twistors [21]. It turns out that the algebra obtained by Penrose's analysis will not be an alternative one which is a necessary condition for split octonions. However, a slight modification in Penrose's treatment can give us the split octonions. As we will see, Penrose's canonically quantized twistor algebra is secretly the "square-root" of the conformal algebra but it is not enough to get the split octonions. To get the split octonions we will have to choose a product that reduces the conformal algebra $\mathfrak{su}(2, 2)$ to the automorphism algebra of the split octonions $\mathfrak{g}'_2$. What is twistor space? What does Penrose mean by quantizing it and why does it give us the conformal algebra? How is it related to the split octonions? In this article, we will try to answer all these questions in the simplest way possible. We begin first by going through Penrose's construction.

6.2 Quantized twistors

Twistors are spinors of the conformal group of complexified spacetime. The conformal group is an extension of the Poincare group and in addition to rotations and translations also causes dilations (scaling) and special conformal transformations (some combination of inversion and rotations). A spacetime where each point in the manifold is associated with a tangent space isomorphic to $\mathbb{C}^4$ is complexified spacetime. One can obtain different real spacetimes by taking different "real slices" of this space. We are interested in Lorentzian space where the conformal group is $SO(2, 4)$ with the double cover $SU(2, 2)$. Spinors of the conformal group transform as the fundamental representation of its double cover $SU(2, 2)$ and hence are represented in the space $\mathbb{C}^4$. Let us denote a twistor as:

$$Z^\alpha = (Z^0, Z^1, Z^2, Z^3) \in \mathbb{T} \cong \mathbb{C}^4. \quad (88)$$

Note that the conformal group for complexified spacetime is $SO(6, \mathbb{C})$ with the double cover $SL(4, \mathbb{C})$. A general twistor in $\mathbb{C}^4$ can transform as a spinor of complexified spacetime but for it to be a Lorentzian twistor it should be equipped with a metric with a split signature. We use the metric to define the following inner-product on twistor space:

$$g(Z, W) = g_{\alpha\beta} Z^\alpha \bar{W}^\beta. \quad (89)$$

The metric $g_{\alpha\beta}$ is given by:

$$g_{\alpha\beta} = \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0. \end{pmatrix} \quad (90)$$

This metric is equivalent to the split-signature metric $\eta_{\alpha\beta} = \text{diag}(1, 1, -1, -1)$ as can be seen from a simple change of coordinates. One notices immediately that the split signature in the metric breaks twistor space into three regions. If the norm of a twistor is the inner product of the twistor with itself then the twistor can either have a positive

($\mathbb{T}^+$), negative ($\mathbb{T}^-$), or zero norm ($\mathbb{N}$). Of all these regions, the third seems to be the most interesting since all the null rays in Minkowski space correspond to a unique null twistor.

To see this we need to introduce the two-spinor notation. Any point in Minkowski spacetime can be described as a 2×2 complex hermitian matrix $h_2[\mathbb{C}]$:

$$x^a \sim x^{AA'} = \frac{\sigma_a^{AA'} x^a}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} t+z & x+iy \\ x-iy & t-z \end{pmatrix}. \quad (91)$$

Here, σ_a , with $a = 0, 1, 2, 3$ are the Pauli matrices with the 2-dimensional identity matrix σ_0 included. In this formalism, a spacetime vector is indicated with a pair of left and right-handed spinor indices. The idea is to see spacetime as a “square-root” of some underlying spinor space. Higher tensors can similarly be written in terms of spinor indices by tensoring vectors and co-vectors.

The determinant of this matrix gives the norm of the spacetime vector.

$$\begin{aligned} \det(x^{AA'}) &= \epsilon_{AB} \epsilon_{A'B'} x^{AA'} x^{BB'} \\ &= t^2 - x^2 - y^2 - z^2. \end{aligned}$$

Therefore, the Levi-Civita tensors $\epsilon_{A'B'}$ and ϵ_{AB} act as metrics on the primed and unprimed spinor spaces respectively. Since the determinant gives the norm of the vectors, all the points in spacetime that are null-like separated from the origin will have a zero determinant. Such matrices can be constructed just by tensoring a spinor $\pi_{A'} \in \mathbb{C}^2$ with its hermitian conjugate:

$$p_{AA'} = \bar{\pi}_A \pi_{A'} = \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix} \begin{pmatrix} x & y \end{pmatrix} = \begin{pmatrix} \bar{x}x & \bar{x}y \\ \bar{y}x & \bar{y}y \end{pmatrix}. \quad (92)$$

Thus, null vectors can be represented just by one spinor $\pi_{A'} \in \mathbb{C}^2$. Physically this can be understood by holding a torch that emits light in all directions. The light rays propagate radially and at any constant time they will lie on a 2-sphere which is isomorphic to $\mathbb{CP}^1$, the projective space of $\mathbb{C}^2$. Projectivizing $\mathbb{C}^2$ is identifying all the 2-spheres at different time slices through an equivalence relation. This includes the sphere formed by these null rays at infinite time called the celestial sphere.

At the heart of twistor theory lies the twistor incidence relation which is a way to construct a twistor by assigning a right-handed spinor to a spacetime point in the following way:

$$Z^\alpha = (ix^{AA'} \pi_{A'}, \pi_{A'}). \quad (93)$$

More abstractly, if we consider the tautological bundle $\mathbb{S}$ of $\mathbb{C}^2$ (right-handed spinors) over $h_2[\mathbb{C}]$ (spacetime) then twistor space is a fiber of this bundle given by

$$(x^{AA'}, \pi_{A'}) \in \mathbb{S} \longrightarrow (ix^{AA'} \pi_{A'}, \pi_{A'}) \in \mathbb{T}. \quad (94)$$

The twistors obtained this way are a quadrupole of complex numbers. So it is no surprise that they transform as spinors of the conformal group. What is non-trivial about the twistor incidence relation is that the conformal transformations of the twistor cause the correct conformal transformations to the associated spacetime point as well.

Since one is interested in conformal transformations defined up to a scaling, projective twistor space is of more interest than twistor space itself. But how to construct a projective twistor space starting from a space that has an isotropic metric? Again the resolution is to consider the projective subspace with positive norms $\mathbb{PT}^+$, projective subspace with negative norms $\mathbb{PT}^-$, and the null projective twistors $\mathbb{PN}$. Projective twistor space $\mathbb{PT}$ is then given by:

$$\mathbb{PT} = \mathbb{PT}^+ \cup \mathbb{PT}^- \cup \mathbb{PN}. \quad (95)$$

We will identify the projective twistor space $\mathbb{PT}$ with $\mathbb{CP}^3$. $SU(2, 2)$ will have open orbits in $\mathbb{PT}^+$, $\mathbb{PT}^-$, and $\mathbb{PN}$. Note that twistor space is a pseudo-Kahler manifold (in fact they belong to an even more special class of complex manifolds called the Calabi-Yau manifold) and comes with a complex structure “ i ”. Using the metric and the complex structure we can define a symplectic 2-form on this manifold:

$$\Omega(Z, \bar{Z}) = \eta(iZ, \bar{Z}). \quad (96)$$

In fact, in a Kahler manifold one can define either the metric, the complex structure, or the symplectic 2-form in terms of the remaining two quantities. More explicitly we can write:

$$\Omega_{\alpha\beta} = i\eta_{\alpha\beta} \quad (97)$$

$$\Omega = \Omega_{\alpha\beta} dZ^\alpha \wedge d\bar{Z}^\beta. \quad (98)$$

Further, this symplectic 2-form can be inverted using the metric separately on $\mathbb{PT}^+$ and $\mathbb{PT}^-$ to give a bi-vector. This bi-vector acts as the Poisson bracket for twistor phase space. The bi-vector is given by:

$$\{-, -\} = -i \left(\frac{\partial}{\partial Z^\alpha} \frac{\partial}{\partial \bar{Z}_\alpha} - \frac{\partial}{\partial \bar{Z}_\alpha} \frac{\partial}{\partial Z^\alpha} \right). \quad (99)$$

To canonically quantize the twistor space we will use the good-old trick used by physicists which involves upgrading the phase space elements to operators acting on a Hilbert space and replacing the Poisson bracket with a commutation relation that defines the operator algebra acting on the Hilbert space. This must be done carefully on $\mathbb{PT}^+$ and $\mathbb{PT}^-$ separately due to the minus signs in the metric. We make the following changes to quantize twistor phase space:

$$Z^\alpha \longrightarrow \frac{\partial}{\partial \bar{Z}_\alpha}, \quad \bar{Z}_\alpha \longrightarrow -\frac{\partial}{\partial Z^\alpha} \quad (100)$$

$$[Z^\alpha, \bar{Z}_\beta] = \delta_\beta^\alpha, \quad [Z^\alpha, Z^\beta] = 0, \quad [\bar{Z}_\alpha, \bar{Z}_\beta] = 0. \quad (101)$$

Quantization of twistor phase space has immense physical significance. The canonical commutation relations given in equation (101) give a unitary representation of the conformal algebra. It is not an irreducible representation as $\mathfrak{su}(2, 2)$ commutes with the invariant norm:

$$s = \frac{1}{2} g_{\alpha\beta} Z^\alpha \bar{Z}^\beta = \frac{1}{2} Z^\alpha \bar{Z}_\alpha. \quad (102)$$

However, if we centrally extend the conformal algebra with s then equation (101) does give us an irreducible unitary representation of the conformal group. This can be seen most

easily if we write the generators of the conformal algebra as a pair of twistor variables. The conformal algebra relations will then emerge naturally from the quantized twistor algebra which can be seen as a “square-root” of the conformal algebra. Let us consider the twistor $Z^\alpha = (\omega^A, \pi_{A'}) = (ix^{AA'}, \pi_{A'})$. The quantized twistor algebra (only the equations that evaluate to a non-trivial value) in terms of the $\omega^A, \pi_{A'}$ variables are:

$$[\omega^A, \bar{\pi}_B] = \delta_B^A, \quad [\pi_{A'}, \bar{\omega}^{B'}] = \delta_{A'}^{B'}. \quad (103)$$

The generators for the conformal algebra and their corresponding operators after quantization are listed below:

- $p_{AA'} = i \frac{\partial}{\partial x^{AA'}} = -\pi_{A'} \frac{\partial}{\partial \omega^A} \longrightarrow \pi_{A'} \bar{\pi}_A$ (translations),
- $M_{AA'BB'} = x_{AA'} \frac{\partial}{\partial x^{BB'}} - x_{BB'} \frac{\partial}{\partial x^{AA'}} \longrightarrow i\epsilon_{A'B'} \pi_{(A} \bar{\omega}_{B)} - i\epsilon_{AB} \omega_{(A'} \bar{\pi}_{B')}$ (rotations + boosts),
- $k_{AA'} = -i\omega_{A'} \frac{\partial}{\partial \pi^A} \longrightarrow \omega_{A'} \bar{\omega}_A$ (special conformal transformations)
- $D = \frac{i}{2} (\pi_A \frac{\partial}{\partial \pi^A} - \omega_{A'} \frac{\partial}{\partial \omega^{A'}}) \longrightarrow -\frac{i}{2} (\pi_A \bar{\omega}^A - \omega_{A'} \bar{\pi}^{A'})$ (dilations).

The Lie algebra of the conformal group that is satisfied by the generators written above comes very neatly from the quantized twistor algebra given in equation(101).

We still haven't described the Hilbert space on which these operators act! The quantized twistor relations give us a way of replacing the anti-holomorphic coordinate $\bar{Z}_\alpha$ with the holomorphic partial derivative $-\frac{\partial}{\partial Z^\alpha}$. Thus the Hilbert space on which these operators act is given by the space of holomorphic functions on twistor space.

Note that the momentum generator is a null vector, so far we are considering only massless particles. This is intrinsic to twistor theory which seems to work best for massless particles. One can of course take a linear superposition of null vectors to get a time-like vector and describe mass. Other than mass, spin is also an invariant of the Poincare algebra and it can be obtained from the Pauli-Lubanski vector that can be obtained by a 4-D cross product between momentum $p_{AA'}$ and angular momentum $M_{AA'BB'}$:

$$S^a = \frac{1}{2} \epsilon^{abcd} M_{bc} p_d \quad (104)$$

$$S^a = s p^a. \quad (105)$$

Upon calculating, one can show that the spin of a particle in terms of the quantized twistors is given by:

$$s = \frac{1}{4} (Z^\alpha \bar{Z}_\alpha + \bar{Z}_\alpha Z^\alpha). \quad (106)$$

Using the commutation relations, one finds:

$$-2s - 2 = Z^\alpha \frac{\partial}{\partial Z^\alpha}. \quad (107)$$

This is the homogeneity operator, giving the order of homogeneity of a polynomial in the twistor variable Z^α . Therefore a particle with spin s can be written in the twistor

Hilbert space in terms of a homogeneous holomorphic function of order $-2s - 2$. This result is known as the Penrose transform and can be used to write classical on-shell fields in spacetime as a contour integral of these functions in twistor space.

It is essential to make the distinction between first-quantization and second-quantization in the twistor language. First quantization is basically what we have described here, and is essentially what Penrose needs to get the split-octonions. First quantization gives us the conformal algebra and the on-shell fields. However, second quantization involves quantizing the off-shell fields on twistor space that need not correspond to holomorphic functions. The way to do this is to write the path integral action for quantum fields on twistor space as has been described by authors in [22, 23].

6.3 Penrose's construction

Penrose describes an 8-dimensional vector space spanned by the quantized twistor Z^α and its dual twistor $\bar{Z}_\alpha$. Any element $\mathcal{A}$ in this bitwistor space can be written as:

$$\mathcal{A} = \bar{A}_\alpha Z^\alpha + A^\alpha \bar{Z}_\alpha \in \mathbb{BT}, \quad (108)$$

for some coefficient $A^\alpha \in \mathbb{C}^4$.

This 8-D real vector space comes from the 4-D complex space and therefore, in the next step Penrose retrieves the complex structure that acts as an endomorphism on this 8-D real space. The complex structure is defined to be the linear map $\mathcal{I} : \mathbb{BT} \rightarrow \mathbb{BT}$ that squares to -1 given by:

$$\mathcal{I}\mathcal{A} = i\bar{A}_\alpha Z^\alpha - iA^\alpha \bar{Z}_\alpha. \quad (109)$$

Now we have the complex structure and the symplectic 2-form which describes the commutation relations. Therefore, using the property of Kahler manifolds we can re-construct our metric but this time on an 8-D real manifold where it will have a $(4, 4)$ split signature. The inner-product between any two bitwistors $\mathcal{A}$ and $\mathcal{B}$ is given by:

$$\mathcal{A} \cdot \mathcal{B} = i[\mathcal{A}, \mathcal{I}\mathcal{B}], \quad (110)$$

$$= A_\alpha B^\alpha + A^\alpha B_\alpha. \quad (111)$$

It is interesting to note that because of the commutation relations between Z^α and $\bar{Z}_{\alpha\text{pha}}$, the terms dependent on them vanish in this inner-product and we are left with just a real number.

Further, Penrose defines a triple product from the bitwistor space to itself $[_, _, _] : \mathbb{BT} \times \mathbb{BT} \times \mathbb{BT} \rightarrow \mathbb{BT}$. This cross-product is defined just using the complex structure and the commutation relations:

$$[\mathcal{A}, \mathcal{B}, \mathcal{C}] = \sum_{\odot} i\mathcal{A}[\mathcal{B}, \mathcal{C}] \quad (112)$$

$$= \sum_{\odot} i\mathcal{A}(\bar{B}_\beta C^\beta - \bar{C}_\beta B^\beta), \quad (113)$$

where the sum is over cyclic permutations of $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$. Just like the inner-product, the dependence on Z^α and $\bar{Z}_\alpha$ vanishes because of the commutation relations.

Split octonions are an unital composition algebra, so they have an identity element in the vector space. Therefore, the next step in Penrose's construction involves choosing an identity element in the bitwistor space. This is an arbitrary choice and possibly breaks the so far unbroken conformal symmetry. Let us choose an identity bitwistor:

$$\mathcal{E} = E_\alpha Z^\alpha + E^\alpha \bar{Z}_\alpha, \quad (114)$$

such that

$$\mathcal{E} \cdot \mathcal{E} = 2 \quad \Leftrightarrow \quad E_\alpha E^\alpha = 1. \quad (115)$$

Once we have the identity vector we need to construct a product on this vector space to make an algebra. This must be done in such a way that the product of any element in the algebra with the identity is itself. Penrose constructs a geometric algebra-inspired product on this vector space. First, we must break the component of any element in the algebra along (scalar component) and perpendicular to the identity (vector component). The scalar and vector part of an element $\mathcal{A}$ in the algebra is given by:

$$\mathcal{A}^S = \frac{1}{2} \mathcal{A} \cdot \mathcal{E}, \quad (116)$$

$$\mathcal{A}^V = \mathcal{A} - \mathcal{A}^S \mathcal{E}. \quad (117)$$

Once we have all the ingredients, we can define the product between any two elements of the algebra as:

$$\begin{aligned} \mathcal{A} \vee \mathcal{B} &= (\mathcal{A}^S \mathcal{E} + \mathcal{A}^V) \vee (\mathcal{B}^S \mathcal{E} + \mathcal{B}^V) \\ &= (\mathcal{A}^S \mathcal{B}^S - \mathcal{A}^V \cdot \mathcal{B}^V) \mathcal{E} + (\mathcal{A}^S \mathcal{B}^V + \mathcal{B}^S \mathcal{A}^V + \mathcal{A}^V \times \mathcal{B}^V) \in \mathbb{BT}. \end{aligned} \quad (118)$$

Where the cross-product between any two elements of the algebra is defined using the triple product on the bitwistor space, the complex structure, and the identity vector:

$$\mathcal{A} \times \mathcal{B} = [\mathcal{I}\mathcal{A}, \mathcal{I}\mathcal{B}, \mathcal{I}\mathcal{E}]. \quad (119)$$

The cross-product is defined in such a way as to ensure that it is orthogonal to both its arguments and the identity vector. In [21] it was claimed that the algebra obtained this way on the 8-D real bitwistor space is the algebra of the split octonions. However, it seems that this algebra is not alternative and is not a composition algebra. An alternative algebra is one where the associator

$$[a, b, c] = (ab)c - a(bc) \quad (120)$$

is completely anti-symmetric in its arguments. From Cayley-Dickson construction and Hurwitz's theorem, it can be concluded that there are only seven composition algebras (over the field of real numbers), the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$, the quaternions $\mathbb{H}$, the octonions $\mathbb{O}$, and their split versions: split complex numbers $\mathbb{C}'$, split quaternions $\mathbb{H}'$, and the split octonions $\mathbb{O}'$. Artin showed that all composition algebras are also alternative [8]. It can be checked that the algebra constructed from the bitwistor space is not a composition algebra either. A composition algebra should satisfy:

$$(\mathcal{X} \wedge \mathcal{Y}) \cdot (\mathcal{X} \wedge \mathcal{Y}) = (\mathcal{X} \cdot \mathcal{X})(\mathcal{Y} \cdot \mathcal{Y}) \quad \forall \mathcal{X}, \mathcal{Y} \in \mathbb{BT}. \quad (121)$$

Penrose's algebra is very interesting because everything is constructed purely out of the twistor commutation relations that seem to respect the conformal algebra. Choosing the

identity seems like the only step that might break the conformal symmetry. However, it is not the algebra of split octonions, the split octonions algebra should respect G_2^* symmetry which is the automorphism group of the split octonions.

What goes wrong in Penrose's construction? One can construct an octonionic algebra inspired by a geometric algebra product as in equation (118). But as shown in [24], the cross-product should satisfy two conditions:

- $(\mathcal{A} \times \mathcal{B}) \cdot \mathcal{A} = (\mathcal{A} \times \mathcal{B}) \cdot \mathcal{B} = 0$ (Orthogonality)
- $|\mathcal{A} \times \mathcal{B}|^2 = |\mathcal{A}|^2 |\mathcal{B}|^2 - (\mathcal{A} \cdot \mathcal{B})^2$ (Pythagorean law).

The cross-product in equation (119) satisfies the orthogonality condition but it does not satisfy the Pythagorean law. Special cross-products satisfying the above properties exist only in 1, 3, and 7 dimensions corresponding again to the composition algebras. We will need a special kind of 3-form in $\mathbb{R}^7$ to construct the desired cross-product. This 3-form can be obtained from a special 4-form in $\mathbb{R}^8$ called the Cayley form. In the upcoming section, we describe how a split Cayley form can be used to get the split octonions starting from $\mathbb{R}^8$ following which we describe a way to get split octonions from the complex twistor space $\mathbb{C}^4$.

6.4 Split Cayley forms

Split Cayley form is a special 4-form in $\mathbb{R}^8$ compatible with a split signature metric that is stabilized by the group $Spin(3, 4)$. Therefore the space of split Cayley forms can be given by

$$\mathcal{C} = Gl(8, \mathbb{R})/Spin(3, 4) \subset \Lambda^4(\mathbb{R}^8). \quad (122)$$

A split Cayley form uniquely chooses a split-signature metric and a real unit Weyl spinor in the spinor bundle S^+ of $\mathbb{R}^8$. The dimension of $\mathcal{C}$ is given by $\dim(\mathcal{C}) = 64 - 21 = 36 + 7$, where 36 is the space of all split signature metrics on $\mathbb{R}^8$ and 7 is the dimension of a real unit spinor. For example if we have a split Cayley form W_{abcd} on $\mathbb{R}^8$, then it chooses a split metric g^W and a Weyl spinor $\psi^W \in S^+$ such that:

$$\langle \psi^W, \gamma_a \gamma_b \gamma_c \gamma_d \psi^W \rangle = W_{abcd} \quad (123)$$

where $\gamma_i \in Cl(4, 4) \cong \text{End}(S)$ are the generators of the Clifford algebra determined by the metric

$$\{\gamma_a, \gamma_b\} = g_{ab}^W \quad (124)$$

and $\langle, \rangle$ is a $Spin(4, 4)$ invariant inner-product on S^+ .

Analogously for Euclidean spaces there exist the more well-known Cayley forms that we are not introducing here. To get the algebra of octonions from $\mathbb{R}^8$ all we need is a Cayley form! Similarly to get the algebra of split octonions from $\mathbb{R}^8$ all we need is a split Cayley form. Split Cayley forms are calibrations in $\mathbb{R}^8$. A calibration is a four form $W \in \Lambda^4(\mathbb{R}^8)$ such that $\frac{1}{8}(1 + W)$ is an idempotent of the Clifford algebra:

$$\frac{1}{64}(1 + W)^2 = \frac{1}{8}(1 + W) \quad (125)$$

where the product is the Clifford product between forms in $Cl(4, 4)$ given by $ab = a \wedge b + a \lrcorner b$.

The Cayley form defines a triple product on the co-tangent space in $\mathbb{R}^8$. The triple product is map $W : \Lambda^1(\mathbb{R}^8) \times \Lambda^1(\mathbb{R}^8) \times \Lambda^1(\mathbb{R}^8) \rightarrow \Lambda^1(\mathbb{R}^8)$ given by:

$$a \times b \times c := W(a, b, c) = W_{\perp} a_{\perp} b_{\perp} c \quad \text{where } a, b, c \in \Lambda^1(\mathbb{R}^8). \quad (126)$$

This is the desired triple product to get the (split) octonionic algebra and should be the replacement for the triple product defined by Penrose in equation(112). This triple product has the following nice property:

$$|a \times b \times c|^2 = \det \begin{pmatrix} |a|^2 & a_{\perp} b & a_{\perp} c \\ b_{\perp} a & |b|^2 & b_{\perp} c \\ c_{\perp} a & c_{\perp} b & |c|^2 \end{pmatrix}. \quad (127)$$

The calibration (or the split Cayley form) also fixes a direction in $\mathbb{R}^8$. We can denote this direction by a one-form $n \in \Lambda^1(\mathbb{R}^8)$ that satisfies $nW = Wn$. This special direction plays the role of identity in the octonionic algebra and one can now define a cross-product to be a map that takes two one-forms and gives another one-form $\times : \Lambda^1(\mathbb{R}^8) \times \Lambda^1(\mathbb{R}^8) \rightarrow \Lambda^1(\mathbb{R}^8)$:

$$a \times b := W_{\perp} a_{\perp} b_{\perp} n \quad \text{where } a, b \in \Lambda^1(\mathbb{R}^8). \quad (128)$$

Or equivalently, one can define a three-form $V = n_{\perp} W$ such that the cross-product is given by:

$$a \times b = -a_{\perp} b_{\perp} V. \quad (129)$$

Finally, we can break any vector $a \in \mathbb{R}^8$ into components along and perpendicular to n . $a = (\alpha, \vec{a})$ where $\alpha = a_{\perp} n$ and $\vec{a}_{\perp} n = 0$. Now one can write the geometric algebra inspired product on $\Lambda^1(\mathbb{R}^8)$ that defines the split-octonionic algebra:

$$ab = (\alpha\beta - \vec{a}_{\perp} \vec{b})n + (\alpha \vec{b} + \beta \vec{a} + \vec{a} \times \vec{b}). \quad (130)$$

This product defines the split octonions as constructed from one-forms on $\mathbb{R}^8$. The cross-product is orthogonal to the identity one-form n as well as its arguments. Moreover this cross-product satisfies the Pythagorean law which follows from equation(127).

Twistor space $\mathbb{C}^4$ can be obtained by choosing a complex structure in $\mathbb{R}^8$ equipped with a split metric. In the next section we will discuss how to obtain split octonions from twistor space. In [25], Krasnov defines a split complex Cayley form that is the space of 4-forms in $\mathbb{R}^8$ stabilized by $SU(2, 2)$ rather than $Spin(3, 4)$. A complex split Cayley form chooses a unique split signature metric on $\mathbb{R}^8$ and a complex unit Weyl spinor on its spinor bundle. The complex Cayley form that Krasnov describes is most likely related to the construction of split octonions from twistor space described below.

6.5 Split octonions from twistor space

We discuss the construction of split octonions from $\mathbb{C}^4$ equipped with a split metric $\eta_{\alpha\beta}$. This construction is analogous to the construction of octonions from $\mathbb{C}^4$ with a Euclidean metric as described in [26, 27].

Using the split metric we first construct a sesqui-linear inner-product on $\mathbb{C}^4$:

$$Z \cdot W = \bar{Z}^{\alpha} \eta_{\alpha\beta} W^{\beta} \in \mathbb{C}. \quad (131)$$

Just like the previous constructions, we need to choose an identity twistor and decompose the space in directions along and perpendicular to this twistor. Let us choose a twistor $E^\alpha \in \mathbb{C}^4$ such that:

$$E \cdot E = 1. \quad (132)$$

We can now break any twistor into its scalar and vector part $Z = (z, \vec{Z}) \in \mathbb{C} \oplus \mathbb{C}^3$ and construct a $SU(1, 2)$ invariant cross-product on the vector part $\vec{\times} : \mathbb{C}^3 \times \mathbb{C}^3 \rightarrow \mathbb{C}^3$:

$$(\vec{Z} \vec{\times} \vec{W})^i = \eta^{ii'} \overline{(\epsilon_{i'j'k'} \vec{Z}^{j'} \vec{W}^{k'})}. \quad (133)$$

This cross-product is $SU(1, 2)$ invariant in the sense that for $U \in SU(1, 2)$:

$$U \vec{Z} \vec{\times} U \vec{W} = U(\vec{Z} \vec{\times} \vec{W}). \quad (134)$$

We now have all the ingredients to construct the split octonionic product which is given by:

$$ZW = (zw - \vec{Z} \cdot \vec{W})E + (z\vec{W} + \vec{Z}w + \vec{Z} \vec{\times} \vec{W}). \quad (135)$$

The inner product $\vec{Z} \cdot \vec{W}$ on $\mathbb{C}^3$ is the one induced from the twistor metric. The product $z\vec{W}$ is the scalar product of $\mathbb{C}$ over $\mathbb{C}^3$, however, it is a peculiar product where $z\vec{W} = -\vec{W}z$. This is needed for the imaginary octonions to anti-commute, more details can be checked in [26]. Since we are left multiplying W with Z , we get $\vec{Z}$ on the left of w .

The octonionic conjugation is given by:

$$Z^* = (z, -\vec{Z}) \quad (136)$$

and the norm on the algebra is given by:

$$|Z|^2 = ZZ^* = Z^*Z. \quad (137)$$

It can be verified that the algebra constructed this way is a composition algebra and is alternative. The algebra has a $(4, 4)$ split signature and is the algebra of the split octonions.

Notably the cross-product has $SU(1, 2)$ symmetry and the conformal symmetry $SU(2, 2)$ is broken while choosing the identity twistor in this construction. Since, the conformal symmetry comes from the twistor commutation relations, the cross-product constructed from the commutation relations is unlikely to give us the correct split octonion algebra. However, one can still use the twistor commutation relations to obtain the inner-product.

We conclude that we can obtain the split octonions from a split Cayley form on $\mathbb{R}^8$. We can obtain twistor space by choosing a complex structure and a split signature metric from $\mathbb{R}^8$. To obtain split octonions from twistor space we need to choose an identity twistor and define a product that breaks conformal symmetry, on the other hand one can obtain twistor space from split octonions by choosing a complex structure. We summarize these results in the following diagram.

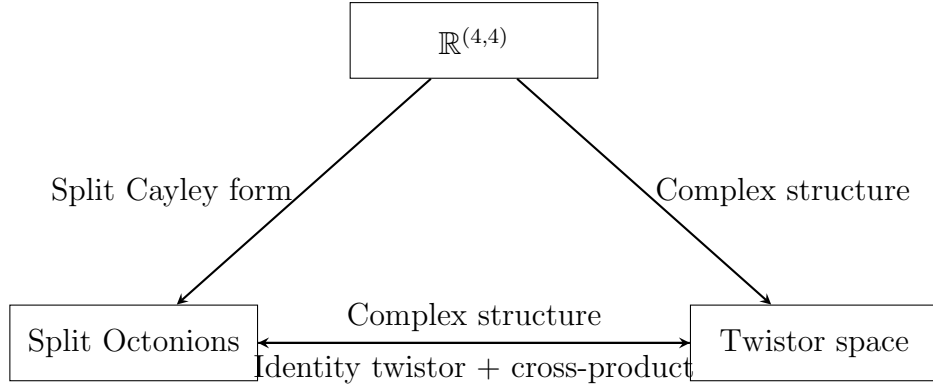


Figure 5: Twistors and Split Octonions

7 Summary

In this project, we have tried to understand the underlying symmetry of the Standard Model using superoctonions. We need to explore many things still, the analysis was in Euclidean signature and we don't know how to get the Lorentzian signature unification. In [28], the authors have shown that the tensor product between split-octonions and octonions $\mathbb{O}' \otimes \mathbb{O}$ transforms as a Weyl-spinor of $\mathfrak{spin}(11, 3)$ that corresponds to $\mathfrak{so}(10)$ internal unification with spacetime $\mathfrak{so}(3, 1)$. However, questions like the Coleman-Mandula theorem remain open in approaches that follow similar paths. In our approach, we have a supersymmetry to deal with the no-go theorem but we haven't been able to find a superalgebra that has a split-octonionic bosonic part $\mathbb{O}'_c$ and an octonionic fermionic part $\mathbb{O}_a$. Having a superalgebra like that would also open an understanding of the $\mathcal{N} = 4$ supertwistors as complexification of $\mathbb{O}'_c \oplus \mathbb{O}_a$. We also need to work more on the higher-derivative gravity couplings and the three-generation claim to ascertain our claims.

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